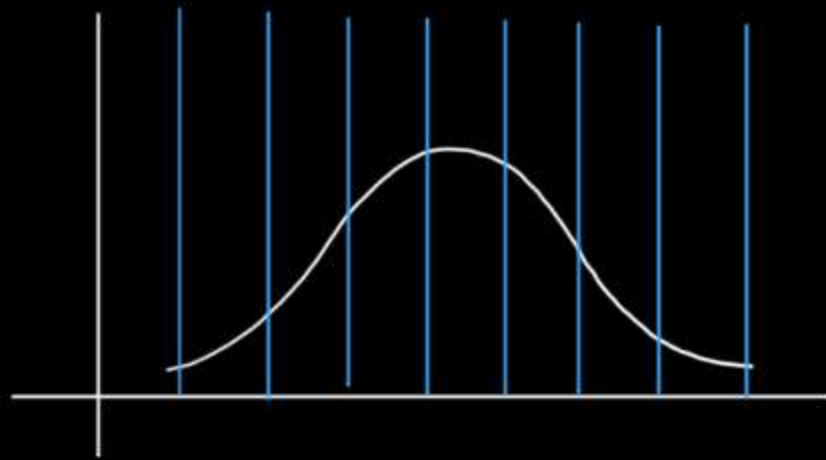


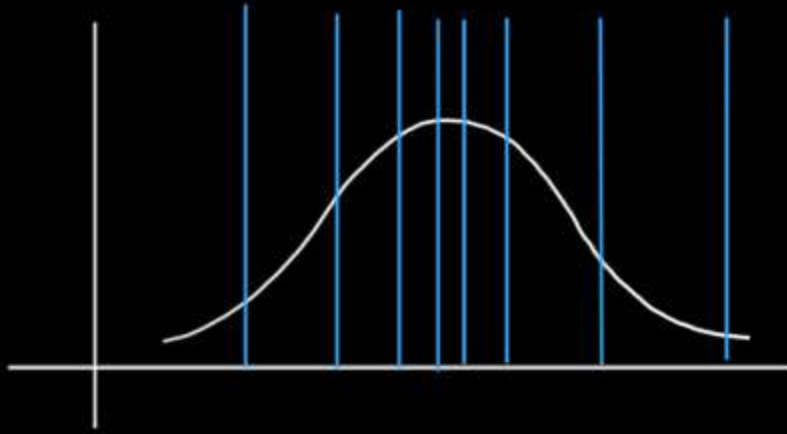
Normal Float Quantization (NF4)

Neural network weights are approximately normally distributed, centered around zero. When quantizing such weight, most values lie near zero, not maximum.

In uniform quantization, equal-sized bins waste precision on rare large magnitude values.



NF4 use non-uniform bins based on percentage of the data. More bins near zero & fewer bins in the tail.



This minimizes expected quantization error for normally distributed weights.